B.TECH. COMPUTER SCIENCE AND ENGINEERING 2022

DATA COMMUNICATION AND NETWORKS

Course Code: 23CSE401 **Credits: 3**
L: T: P: 3:0:0 **Contact Hours: 45**
Prerequisite: Digital Logic Design

Course Objectives

The objective of this course is to make students to

1. Discussion on the various basic concepts and layered network architecture
2. Understanding the concepts related to various network protocols and their association with layers of OSI Reference model
3. Detailed explanation on different types of network addresses, Routing, error detection and correction at different layers of OSI Model
4. Exploring and learning various user level protocols and it's applications

Course Outcomes:

At the end of the course, students will be able to

Course Outcomes	Description	Bloom's Taxonomy Level
CO1	Discuss the Categories and functions of various Data communication Networks	Understanding (2)
CO2	Explain the Design and analyze various error detection techniques.	Understanding (2)
CO3	Demonstrate the mechanism of routing the data in network layer	Applying (3)
CO4	Explain the significance of various Flow control and Congestion control Mechanisms	Understanding (2)
CO5	Discuss the Functioning of various Application layer Protocols.	Understanding (2)

Course Contents

MODULE 1: Introduction to Data Communications **[10 hours]**
Components, Data Representation, Data Flow, Networks- Distributed Processing, Network Criteria, Physical Structures, Network Models, Categories of Networks Interconnection of Networks, The Internet - A Brief History, The Internet Today, Protocol and Standards - Protocols, Standards, Standards Organizations, Internet Standards. Network Models, Layered Tasks, OSI model, Layers in OSI model, TCP/IP Protocol Suite, Addressing Introduction, Wireless Links and

Network Characteristics, WiFi: 802.11 Wireless LANs -The 802.11 Architecture.

MODULE 2: Data Link Layer **[10 hours]**

Links, Access Networks, and LANs- Introduction to the Link Layer, The Services Provided by the Link Layer, Types of errors, Redundancy, Detection vs Correction, Forward error correction Versus Retransmission Error-Detection and Correction Techniques, Parity Checks, Check summing Methods, Cyclic Redundancy Check (CRC) , Framing, Flow Control and Error Control protocols , Noisy less Channels and Noisy Channels, HDLC, Multiple Access Protocols, Random Access ,ALOHA, Controlled access, Channelization Protocols. 802.11 MAC Protocol, IEEE 802.11 Frame.

MODULE 3: The Network Layer **[08 hours]**

Introduction, Forwarding and Routing, Network Service Models, Virtual Circuit and Datagram Networks-Virtual-Circuit Networks, Datagram Networks, Origins of VC and Dat\agram Networks, Inside a Router-Input Processing, Switching, Output Processing, Queuing, The Routing Control Plane, The Internet Protocol(IP):Forwarding and Addressing in the Internet- Datagram format, Ipv4 Addressing, Internet Control Message Protocol(ICMP), IPv6

MODULE 4: Transport Layer **[09 hours]**

Introduction and Transport Layer Services : Relationship Between Transport and Network Layers, Overview of the Transport Layer in the Internet, Multiplexing and Demultiplexing, Connectionless Transport: UDP -UDP Segment Structure, UDP Checksum, Principles of Reliable Data Transfer-Building a Reliable Data Transfer Protocol, Pipelined Reliable Data Transfer Protocols, Go- Back-N(GBN), Selective Repeat(SR), Connection Oriented Transport: TCP - The TCP Connection, TCP Segment Structure, Round-Trip Time Estimation and Timeout, Reliable Data Transfer, Flow Control, TCP Connection Management, Principles of Congestion Control - The Cause and the Costs of Congestion, Approaches to Congestion Control

MODULE 5: Application Layer **[08 hours]**

Principles of Networking Applications – Network Application Architectures, Processes Communicating, Transport Services Available to Applications, Transport Services Provided by the File Transfer: FTP,-FTP Commands and Replies, Electronic Mail in the Internet- STMP, Comparison with HTTP, DNS-The Internet's Directory Service – Service Provided by DNS, Overview of How DNS Works, DNS Records and messages.

TEXTBOOKS:

1. Data Communications and Networking Behrouz A. Forouzan 4th Edition McGraw-Hill Education
2. Computer Networks -- Andrew S Tanenbaum, 4th Edition, Pearson Education

REFERENCES:

1. Data communication and Networks - BhusanTrivedi, Oxford university press, 2016
2. Computer Networking A Top-Down Approach – Kurose James F, Keith W, 6th Edition, Pearson.
3. Data and Computer Communications - William Stallings, 10th Edition, Pearson Education.
4. Understanding Communications and Networks, 3rd Edition, W. A. Shay, Cengage Learning.